

SmartHire Project Report

1. Overview

SmartHire is an AI-Powered Resume-to-Job Matching and Career Guidance Engine. It uses classical Machine Learning models to match uploaded resumes against a corpus of jobs, predict job categories, and identify skill gaps.

2. Architecture & Tech Stack

- Language: Python 3.10+
- Web Framework: Streamlit
- Data Processing: Pandas, NumPy, Scikit-Learn
- NLP & Machine Learning: TF-IDF, Logistic Regression, Cosine Similarity, KMeans

3. Core Components

- Resume Classifier: Supervised model using TF-IDF and Logistic Regression to predict the best job category for a given resume.
- Job Recommender: Unsupervised matching engine using TF-IDF and Cosine Similarity to find top job matches based on skill overlap.
- Skill-Gap Analysis: Compares extracted skills from the resume with required skills from job postings.

4. Workflow

- Data loading and preprocessing from Kaggle datasets.
- Model training via Jupyter notebooks.
- Inference served through an interactive Streamlit frontend.